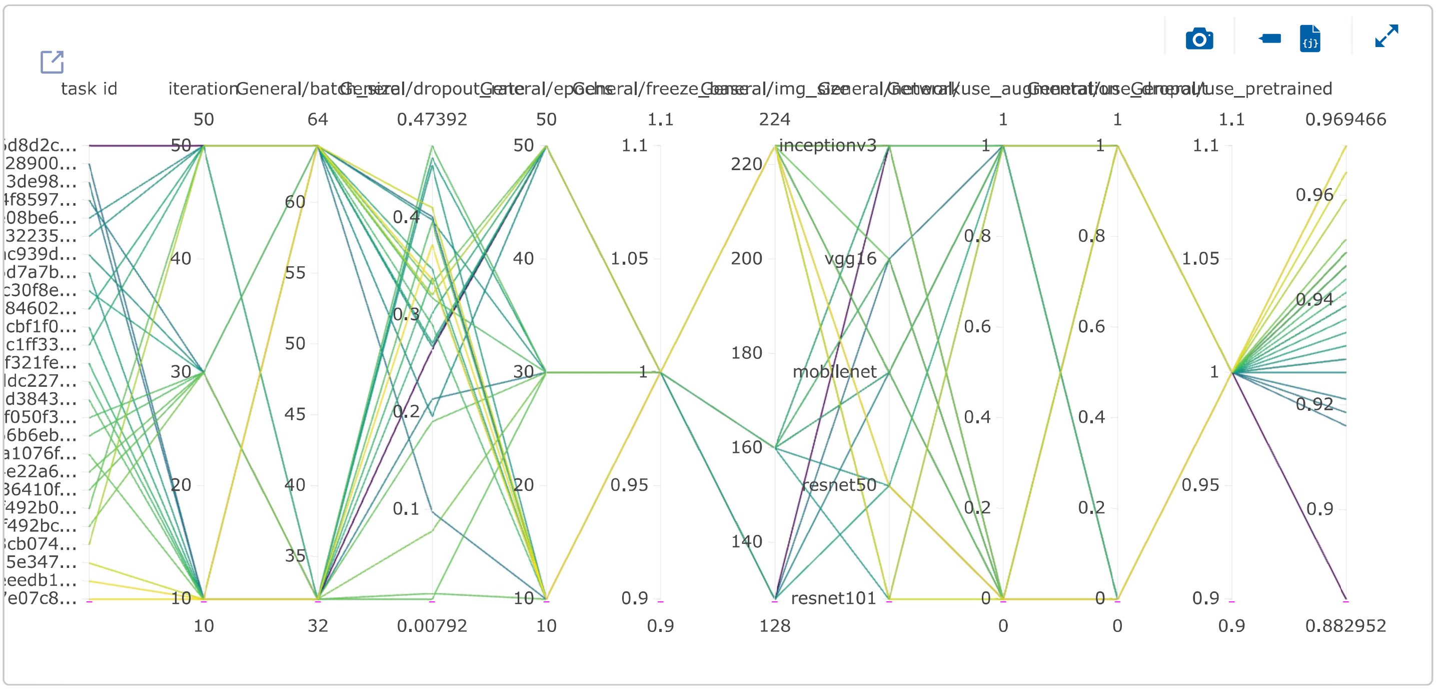
Automatyczna optymalizacja hiperparametrów - full data, transfer

Parametry optymalizatora

```
# Base task
base_task = Task.get_task(project_name="pizza_binary_classification", task_name="hpo_train_task_v7")

# Configuration
optimizer = HyperParameterOptimizer(
    base_task_id=base_task.id,
    hyper_parameters=[
        DiscreteParameterRange("General/network", ["mobilenet", "resnet50", "vgg16", "resnet101", "inceptionv
        DiscreteParameterRange("General/img_size", [128, 160, 224]),
        DiscreteParameterRange("General/use_pretrained", [True]),
        DiscreteParameterRange("General/freeze_base", [True]),
        DiscreteParameterRange("General/use_dropout", [True, False]),
        DiscreteParameterRange("General/use_augmentation", [True, False]),
        UniformParameterRange("General/dropout_rate", 0.0, 0.5),
        DiscreteParameterRange("General/batch_size", [32, 64]),
        DiscreteParameterRange("General/epochs", [10, 30, 50])
    ],
    objective_metric_title="accuracy",
    objective_metric_series="val",
    objective_metric_sign="max",
    execution_queue="default",
    max_number_of_concurrent_tasks=2,
    total_max_jobs=30,
    min_iteration_per_job=3,
    optimizer_type="random_search"
)
```

Parallel Coordinates



summary

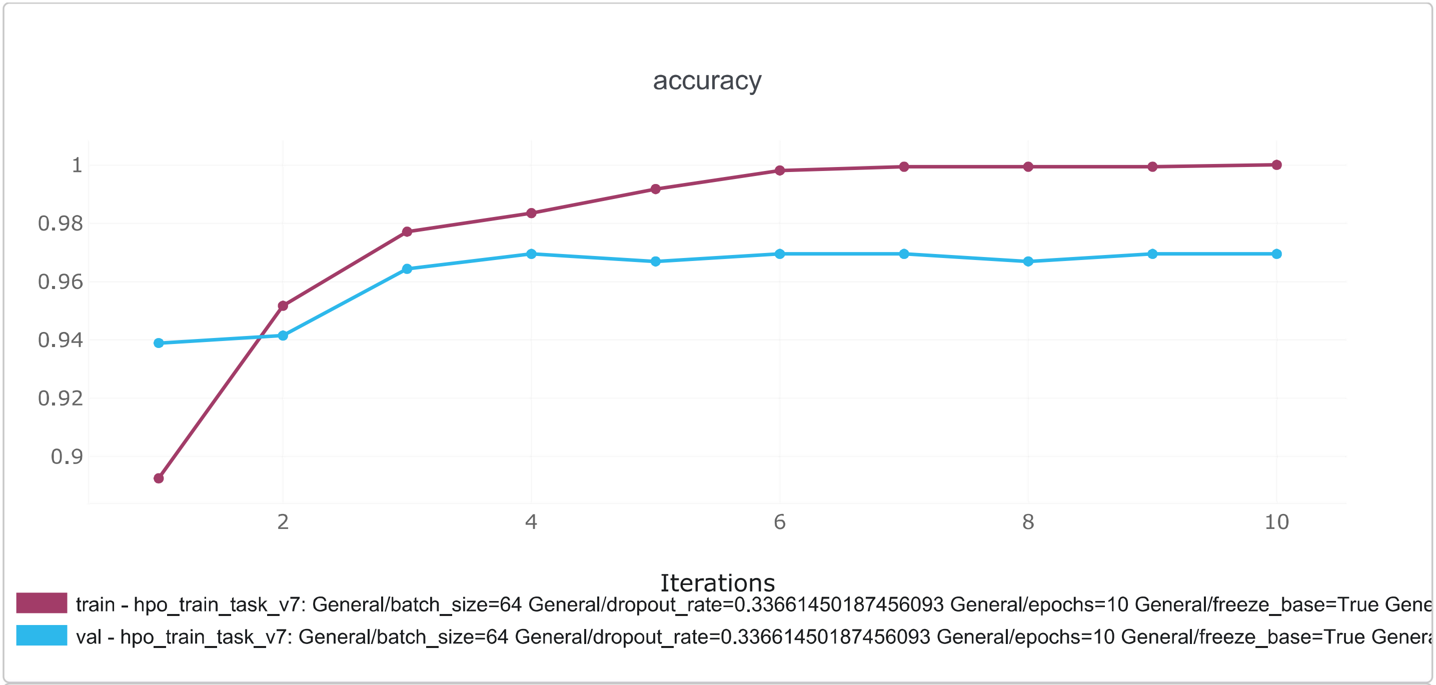
objective: accuracy/val

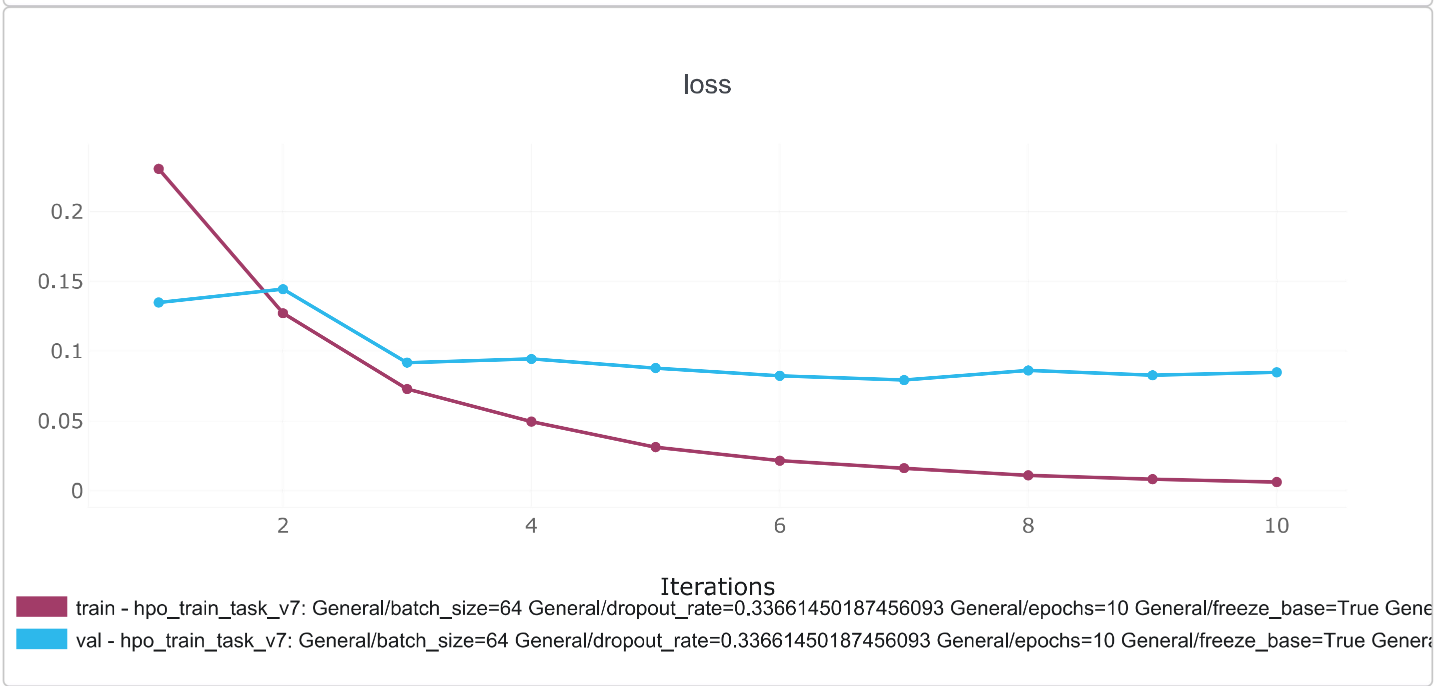
task id	objective	iteration	General/batch_	General/dropo	General/epoch	General/freeze	General/img_s	Gene
7e07c8a961bd	0.9694656729	10	64	0.3366145018	10	true	224	resn
eeedb1a9b526	0.9694656729	10	32	0.3722811795	10	true	224	resn
75e3478dde47	0.9643765687	10	64	0.4102188148	10	true	224	resn
3cb074c7ccec4	0.9592875242	50	64	0.3206530788	50	true	224	resn
f492bc1856ea4	0.9516539573	30	32	0.0779590185	30	true	224	incep
f492b063bd99	0.9491094350	50	64	0.3318744414	50	true	224	incep
36410f54358c	0.9491094350	30	32	0.1902215930	30	true	224	vgg1
4e22a6d4781c	0.9491094350	30	32	0.0079219384	30	true	224	mobi
a1076f04b520	0.9491094350	10	32	0.0138461142	10	true	224	mobi

Zwycięzca - resnet50!!!

Parametr	Wartość
batch_size	64
dropout_rate	0.33661450187456093
epochs	10
freeze_base	True
img_size	224
network	resnet50
use_augmentation	False
use_dropout	False
use_pretrained	True

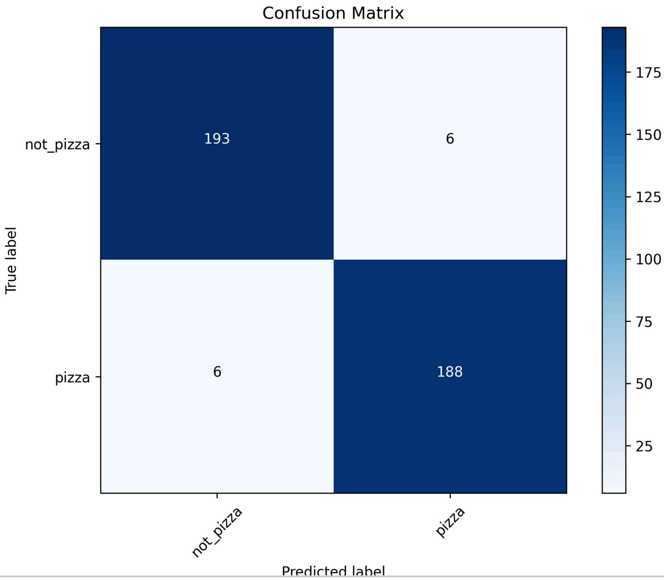
Otrzymane wyniki





Confusion Matrix

Confusion Matrix/plot image - Iteration 512



Training Time: 8 minut 43 sekundy - dwa razy szybciej od modelu bazowego

Accuracy: 96.95% - Hell yeah